

Assignment 4
Kalman Filter and Parameter Estimation

Group 49
Jiayue Che, s252639
Anqi Yu, s254721

Part 1: Parameter Estimation in State-Space Models

1.1 Simulation of Five Realizations

Five independent realizations of the AR(1) process $X_t = aX_{t-1} + b + e_t$ with a = 0.9, b = 1, $\sigma_1 = 1$, $X_0 = 5$ are simulated for n = 100 time steps. The process follows $X_t = aX_{t-1} + b + \sigma_1\varepsilon_t$, where $\varepsilon_t \sim N(0, 1)$. The simulated trajectories are shown in Figure 1.

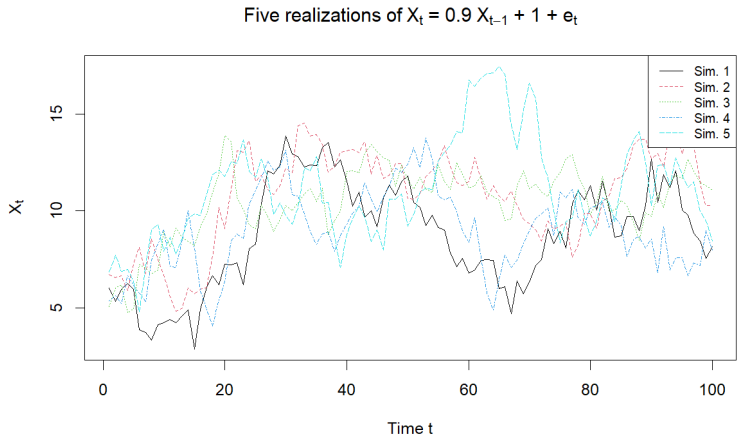


Figure 1. Five simulated realizations of the AR(1) process.

The five trajectories diverge over time due to the stochastic noise, but all fluctuate around the long-run mean b/(1 - a) = 10. The process exhibits strong temporal dependence with a = 0.9.

A random seed was set before simulation to ensure reproducibility of the five trajectories.

1.2 Latent State and Noisy Observations

A single realization is simulated from the state-space model: $X_t = aX_{t-1} + b + \sigma_1 e_{1,t}$, $Y_t = X_t + \sigma_2 e_{2,t}$, where $e_{1,t}, e_{2,t} \sim N(0, 1)$ and $\sigma_1 = \sigma_2 = 1$. The latent state and noisy observations are illustrated in Figure 2.

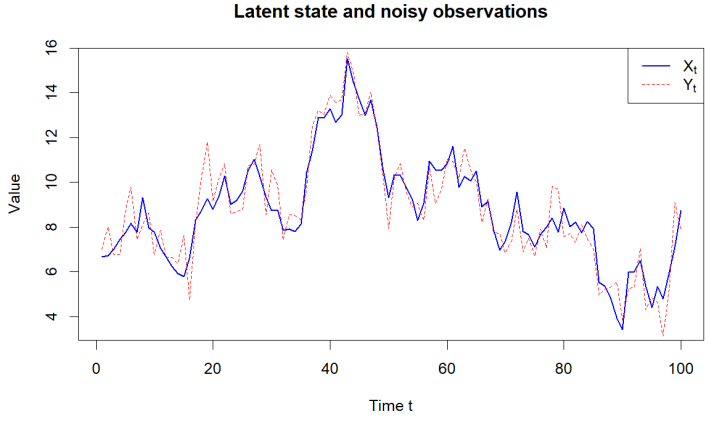


Figure 2. Simulated latent state and corresponding noisy observations.

The observations Y_t track the true state X_t closely but with visible noise. This illustrates the challenge of state-space models: only noisy measurements are available, yet we aim to estimate the parameters of the underlying latent process.

1.3 Kalman Filter Application

The myKalmanFilter function is applied to the simulated data using the true parameters. The filter computes the one-step-ahead predicted state $\hat{X}_{t+1|t}$, its variance $P_{t+1|t}$, the innovation sequence, and the filtered state estimates. A 95% confidence interval is constructed as $\hat{X}_{t+1|t} \pm 1.96\sqrt{P_{t+1|t}}$. The filtering results are presented in Figure 3.

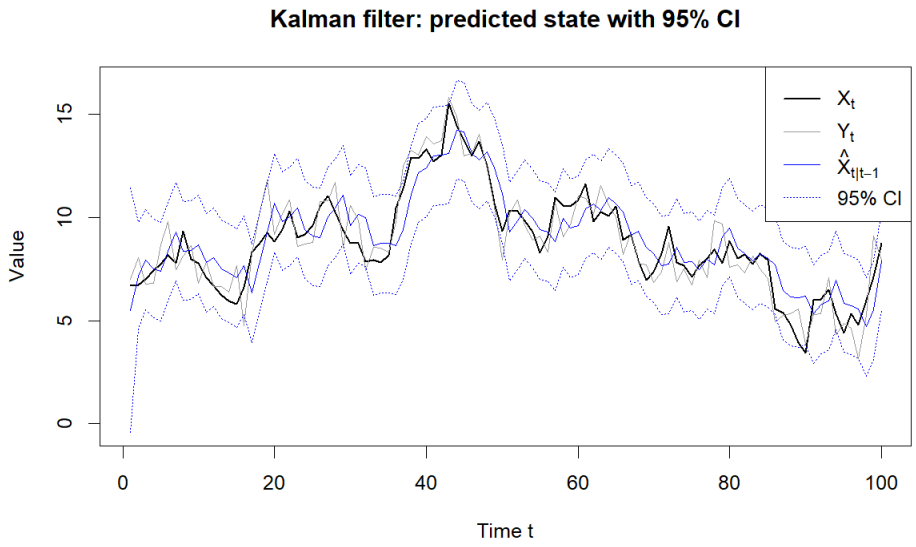


Figure 3. Kalman filter predictions with 95% confidence interval.

The Kalman filter successfully tracks the hidden state. The predicted state follows the true state closely, and the 95% confidence interval covers the true state most of the time. This demonstrates the filter's ability to separate signal from noise even with a simple 1-dimensional model.

1.4 Maximum Likelihood Estimation

Parameters are estimated by minimizing the negative log-likelihood via the L-BFGS-B optimization algorithm. A Monte Carlo study with 100 independent realizations (n = 100 each) is conducted for three parameter combinations: The estimation outcomes are summarized visually in Figure 4.

- Case 1 (Baseline): a = 0.9, b = 1, σ_1 = 1
- Case 2 (Large bias): a = 0.9, b = 5, σ_1 = 1
- Case 3 (Large system noise): a = 0.9, b = 1, σ_1 = 5

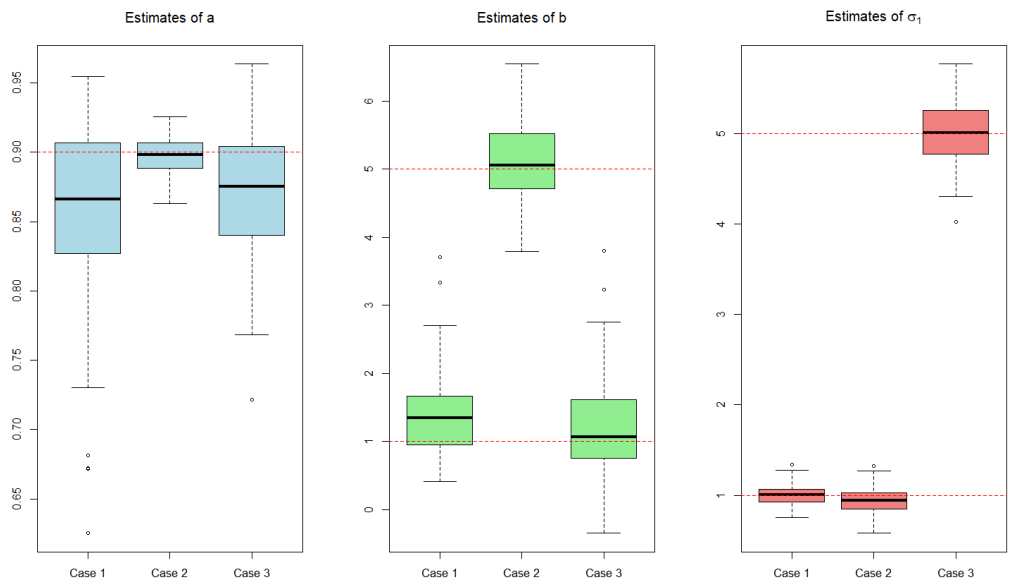


Figure 4. Monte Carlo estimation results for the three parameter cases.

Summary statistics (means) for estimated parameters:

Case	Parameter	Mean
1	a	0.858
1	b	1.405
1	σ_1	1.009
2	a	0.898
2	b	5.112
2	σ_1	0.945
3	a	0.869
3	b	1.189
3	σ_1	5.007

The MLE estimates are generally close to the true values in all three cases. However, some sampling variability is observed, especially for parameters a and b, because the process is highly persistent with a = 0.9. Crucially, the bias parameter b proves more difficult to estimate precisely than the system noise scale σ_1. This is because the strong autoregressive dependence (a = 0.9) induces higher estimation uncertainty and sampling variance for the intercept, whereas σ_1 remains highly identifiable across all cases. In Case 2, where b = 5, the estimates of b are centered close to the true value. In Case 3, where $\sigma_1 = 5$, the estimated system noise scale is also close to the true value, but with larger spread. Overall, the maximum likelihood procedure gives reasonable estimates, although the precision depends on the parameter setting and the amount of system noise.

1.5 Student-t Distributed System Noise

The system noise is now drawn from a scaled Student-t distribution with $\nu \in \{100, 5, 2, 1\}$ degrees of freedom: $X_t = aX_{t-1} + b + \sigma_1 \varepsilon_t$, $\varepsilon_t \sim t(\nu)$.

The Kalman filter still assumes Gaussian noise, allowing us to assess robustness to model misspecification. An example realization under heavy-tailed noise is shown in Figure 5.

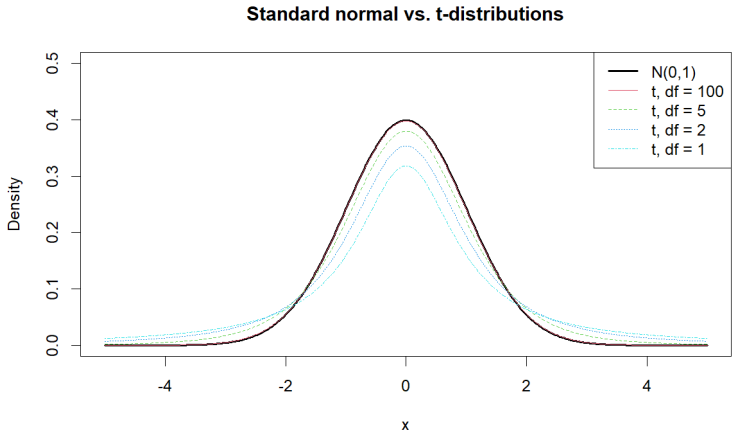


Figure 5. Density comparison between the standard normal distribution and Student-t distributions with different degrees of freedom.

As ν decreases, the t-distribution becomes increasingly heavy-tailed. With $\nu = 2$ the variance is already much larger than the normal, and $\nu = 1$ corresponds to the Cauchy distribution with undefined variance. Heavy-tailed system noise produces more frequent extreme state innovations, which the Gaussian-assuming Kalman filter may misinterpret as evidence of larger system variance. The estimated parameter behavior across different degrees of freedom is displayed in Figure 6.

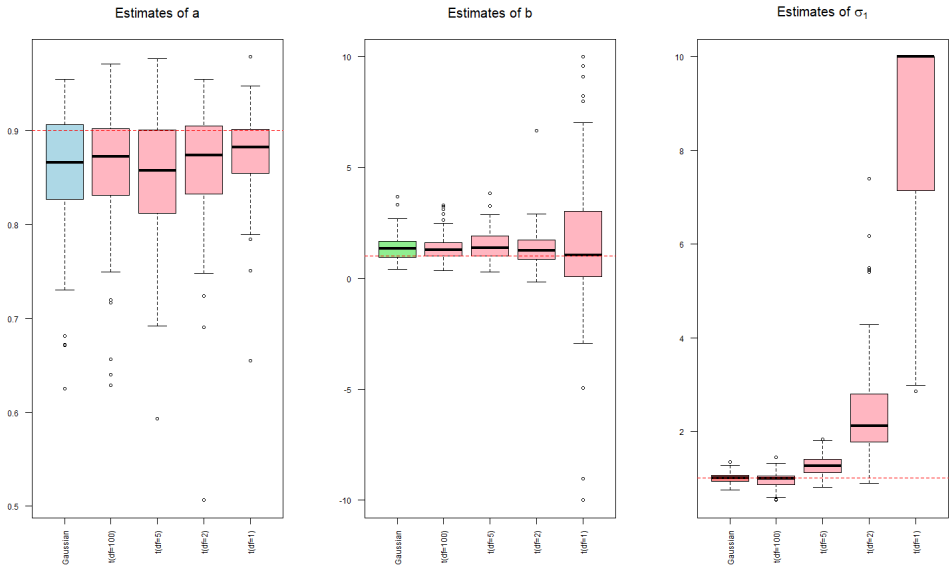


Figure 6. Estimated parameter behavior under varying degrees of freedom.

As ν decreases (heavier tails), the estimates of σ_1 increase to accommodate the more extreme innovations. For ν = 1 (Cauchy), σ_1 is severely overestimated, as the Gaussian likelihood attributes large innovations to high system variance rather than heavy-tailed noise. The estimates of a and b remain relatively stable across different nu, showing partial robustness. This demonstrates that while the Kalman filter remains functional under mild non-Gaussianity, its efficiency and the interpretability of estimated variance parameters degrade as the tails become heavier.

Part 2: Transformer Station Temperature Prediction

2.1 Exploratory Data Analysis

The dataset contains 168 hourly measurements of transformer temperature (Y_t), outdoor air temperature ($T_{a,t}$), solar radiation (S_t), and load current (I_t). The transformer temperature ranges from 19.83 to 38.92 degrees Celsius with a mean of 27.65. Solar radiation shows strong diurnal variation (0 to 755.83 W/m^2), as does the load current (14.39 to 25.36 kA). The exploratory plots are presented in Figure 7.

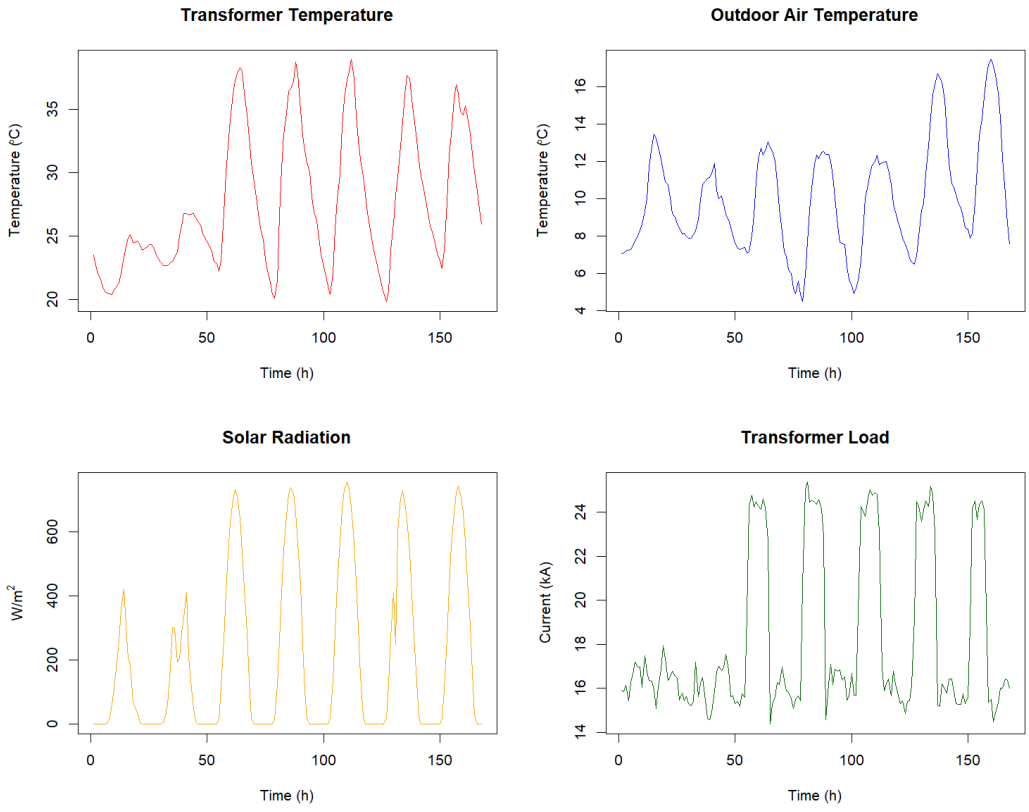


Figure 7. Exploratory plots of transformer temperature and input variables.

The transformer temperature exhibits a pronounced diurnal pattern, peaking during daytime hours when both solar radiation and electrical load are high. Outdoor air temperature follows a similar daily cycle. There is a clear positive relationship between transformer temperature and the three input variables, motivating their inclusion in the state-space model.

2.2 1-Dimensional State-Space Model

A 1-dimensional state-space model of the form $X_{t+1} = A\,X_t + B\,u_t + G\,e_{1,t}$, $Y_t = X_t + e_{2,t}$ is fitted, where $u_t = [T_{a,t}, S_t, I_t]^T$ is the input vector. Seven parameters (A, B_{ta}, B_s, B_i, log Q, log R, X_0) are estimated via maximum likelihood using the Kalman filter.

Estimated parameters:

$A = 0.8009$; $B_{ta} = 0.1020$, $B_s = 0.0029$, $B_i = 0.2134$; $Q = 0.4144$, $R = 0.0001$, $X_0 = 24.2106$

Model fit: log L = -164.56, AIC = 343.12, BIC = 364.99.

The estimated $A = 0.8009$ indicates moderate persistence in the temperature state. The load current has the strongest influence on transformer temperature, with $B_I = 0.2134$, followed by outdoor air temperature, with $B_{Ta} = 0.1020$. Solar radiation has a much smaller direct coefficient, $B_S = 0.0029$, possibly because part of its effect is already reflected through the outdoor air temperature. The very small estimated observation noise, $R_{obs} = 0.0001$, suggests that the model attributes most of the unexplained variation to the state equation rather than to measurement noise.

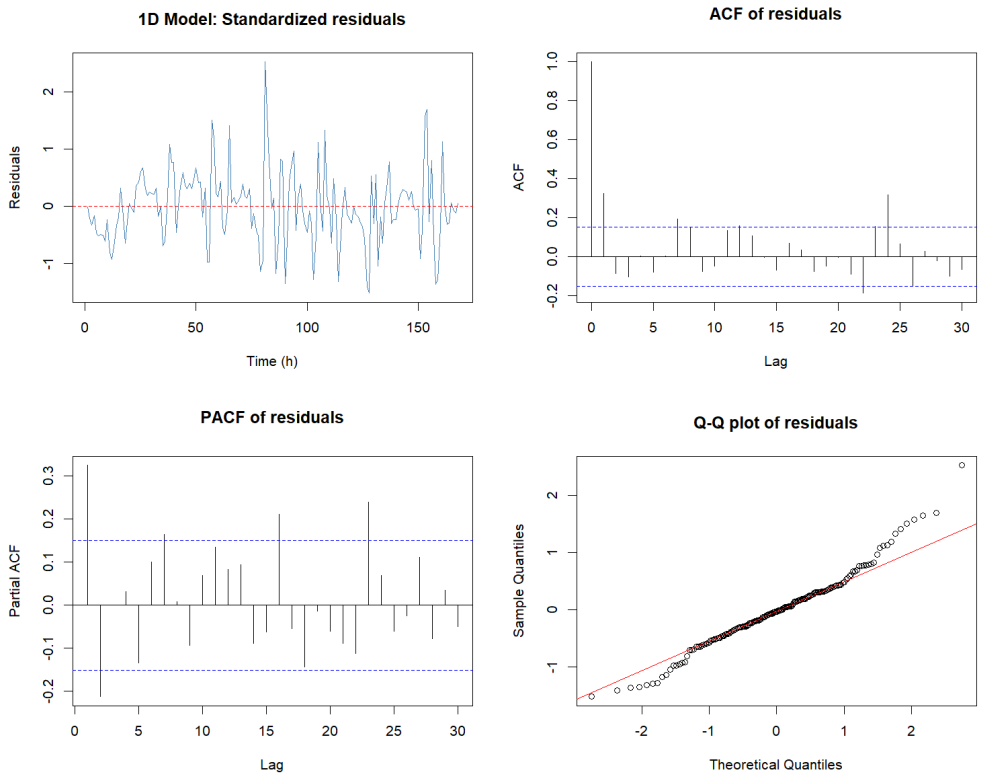


Figure 8. Residual diagnostics for the 1-dimensional state-space model.

The residual ACF shows significant autocorrelation at multiple lags, indicating that a single latent state cannot fully capture the system dynamics. The Q-Q plot suggests approximate normality of the innovations, though with some deviations in the tails.

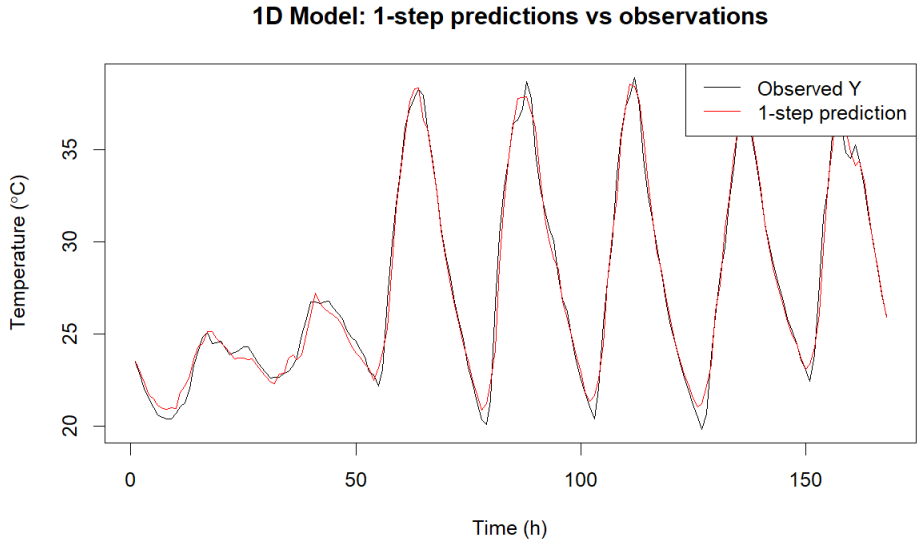


Figure 9. One-step predictions versus observations for the 1-dimensional state-space model.

2.3 2-Dimensional State-Space Model

The model is extended to a 2-dimensional state vector $X_t \in \mathbb{R}^2$. To ensure identifiability, the observation matrix is parameterized as $C = [1, C_2]$ with the first element fixed to 1. A total of 17 parameters are estimated: A (2 × 2), B (2 × 3), the Cholesky factor of the system noise covariance matrix Q (lower-triangular 2×2), C_2 (scalar), the observation noise variance R, and the initial state X_0 (2 × 1). A multi-start optimization strategy (8 starting points initialized from the 1D model solution) is used to avoid local optima. Diagnostic and prediction results for the 2D model are shown in Figures 10 and 11.

Estimated A matrix:

A = [[0.7856, -0.0861],
[0.4712, -0.0197]]

Estimated B matrix:

B = [[0.1075, 0.0020, 0.2416],
[-0.2582, -0.0015, -0.3607]]

Estimated C vector: C = [1, 0.4757]

Estimated R_obs = 0.0151, X_0 = [24.41, 23.25]

Model fit: logL = -124.35, AIC = 282.69, BIC = 335.80.

The 2D model improves the model fit compared with the 1D model. The log-likelihood increases from -164.56 to -124.35, while both AIC and BIC decrease. This suggests that the additional latent state helps explain the temperature dynamics, despite the increased model complexity. The residual autocorrelation is also reduced compared with the 1D model, indicating that the 2D model captures more of the temporal structure in the data.

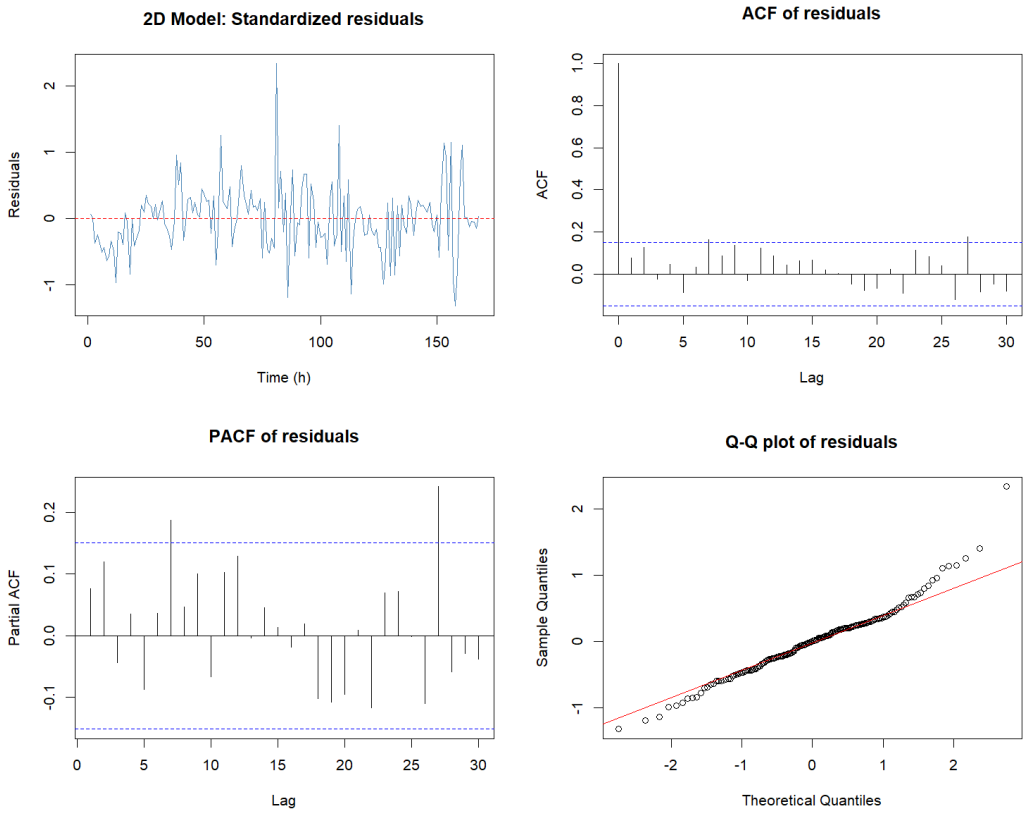


Figure 10. Residual diagnostics for the 2-dimensional state-space model.

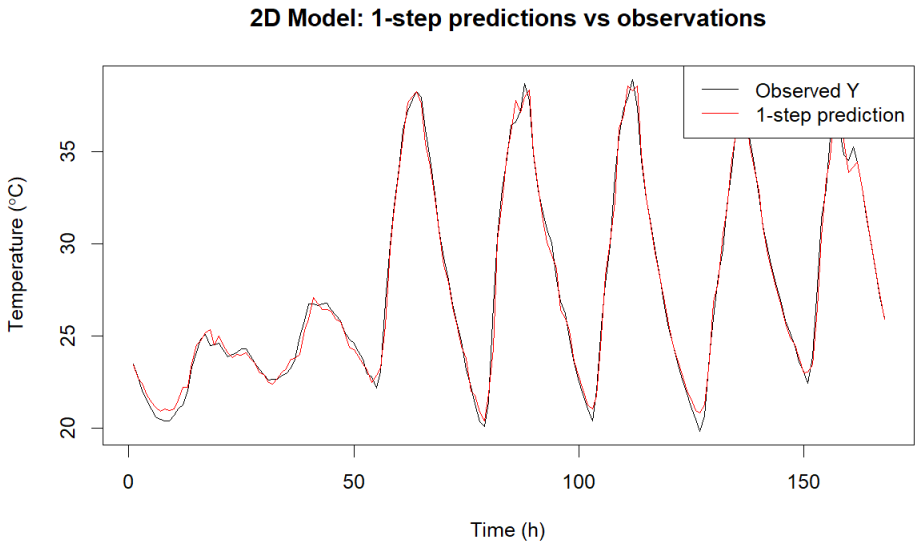


Figure 11. One-step predictions from the 2-dimensional model.

The 2D model innovations show substantially reduced autocorrelation compared to the 1D model. The ACF decays more rapidly with fewer significant lags. The Q-Q plot suggests approximate normality. The 1-step predictions track the observed temperature closely throughout the entire period, including during rapid daytime temperature spikes where the 1D model struggled.

However, despite these substantial mathematical improvements, a critical reflection on the estimated parameter signs reveals a potential physical inconsistency in the model structure. The negative coefficients obtained in the second row of the estimated B matrix imply that increases in outdoor temperature ($T_{a,t}$), solar radiation ($\Phi_{s,t}$), and transformer electrical load ($\Phi_{I,t}$) are associated with decreases in the second latent state. Although the unconstrained maximum likelihood optimization produces a better statistical fit with lower AIC and BIC values, these negative input effects are difficult to interpret directly from a physical perspective. This suggests that the flexibility of the high-dimensional parameterization may introduce parameter collinearity or compensating effects between latent states, allowing the optimization routine to improve the statistical fit without necessarily preserving a fully physically interpretable structure.

2.4 Interpretation of Reconstructed States

The two reconstructed state trajectories are plotted over time, along with the scaled input variables for comparison. The reconstructed state trajectories are presented in Figure 12.

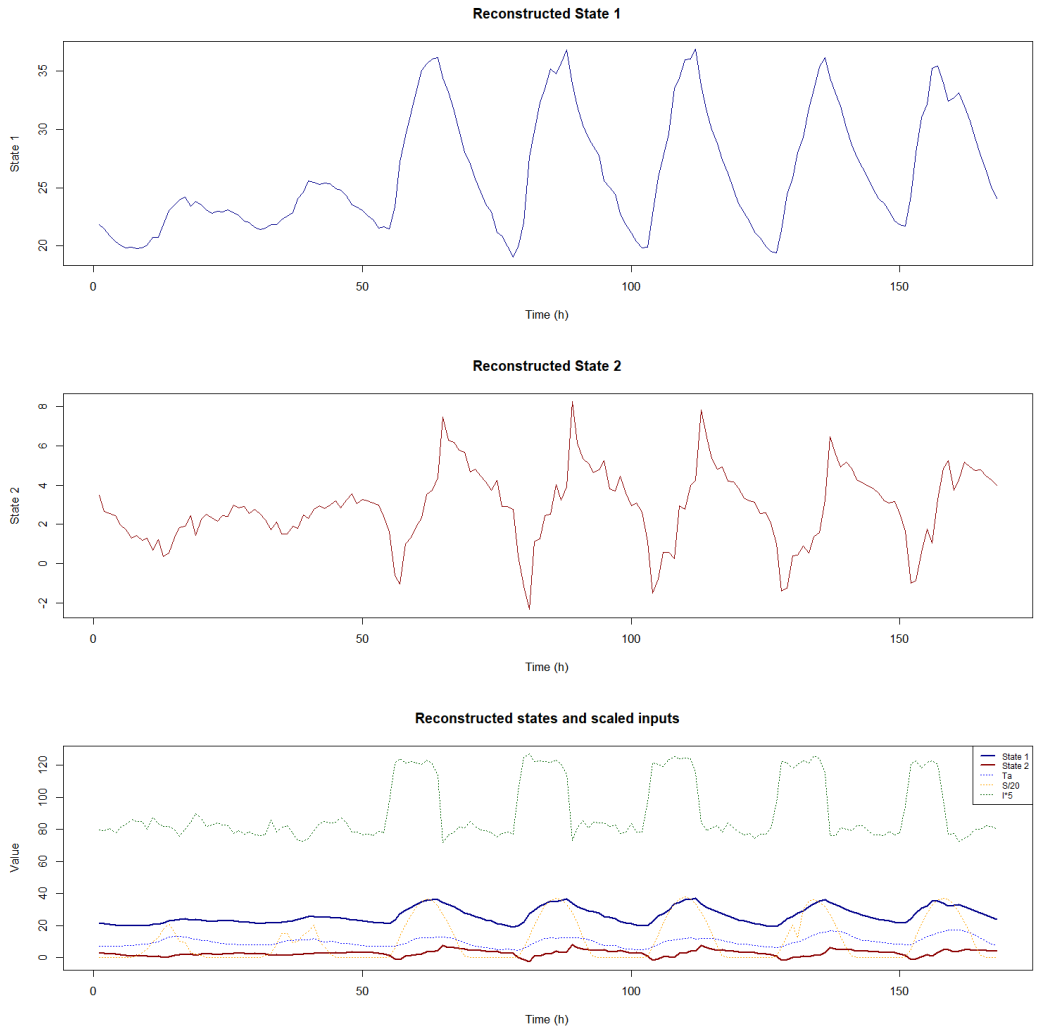


Figure 12. Reconstructed states and scaled input variables.

State 1 can be interpreted as the primary transformer temperature component (the internal copper windings or core). Since $C_1$is fixed to 1, this state directly contributes to the observed temperature. It closely follows the observed temperature pattern and is strongly influenced by the load current ($B_{13} = 0.2416$). This is physically reasonable because higher load current leads to more prominent Joule heating in the transformer windings. The strong self-coupling ($A_{11} = 0.7856$) also perfectly aligns with the system's thermal inertia, indicating that the current core temperature depends heavily on its previous states.

State 2 represents a secondary thermal component, mathematically acting as a delayed cooling or buffering effect, contributing to the observation through $C_2 = 0.4757$. The near-zero self-coupling indicates that this state does not persist strongly by itself, while the massive positive cross-coupling from State 1 ($A_{21} = 0.4712$) proves that it dynamically absorbs heat emitted from the main core.

However, the negative input coefficients for State 2 ($B_{21} = -0.2582$, $B_{23} = -0.3607$) present a severe thermodynamic contradiction. As visualized in our trajectory plots, it forces State 2 to drop close to 0°C during intense daytime heating hours. While this interpretation must be treated with extreme caution, it reveals that the latent state acts less like a physical oil medium and more like an empirical "numerical corrector" that utilizes unconstrained negative scaling to cancel out daytime peak residuals.

In conclusion, while the two-state structure provides an outstanding mathematical fit and a highly flexible description of the temporal dynamics compared to the 1D model, its physical plausibility is highly debatable under unconstrained maximum likelihood optimization. The model achieves an optimal AIC (282.69) by sacrificing thermodynamic realism, treating the secondary state as a mathematical error-sink rather than a literal physical temperature component.